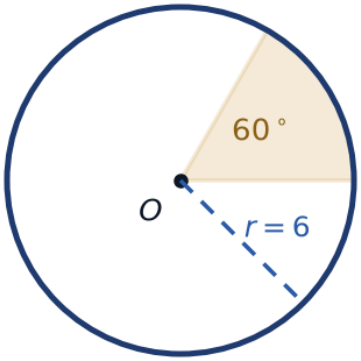
Trigonometry & Polar Coordinates

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

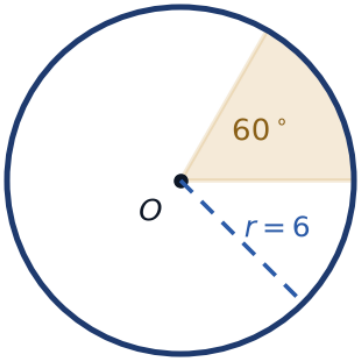
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the polar coordinates $\left(6, \frac{5\pi}{4}\right)$ into rectangular Cartesian coordinates (x, y).



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the Cartesian equation $x^2 + (y - 4)^2 = 16$ into a polar equation relating r and θ.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{\tan(2x)}$

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

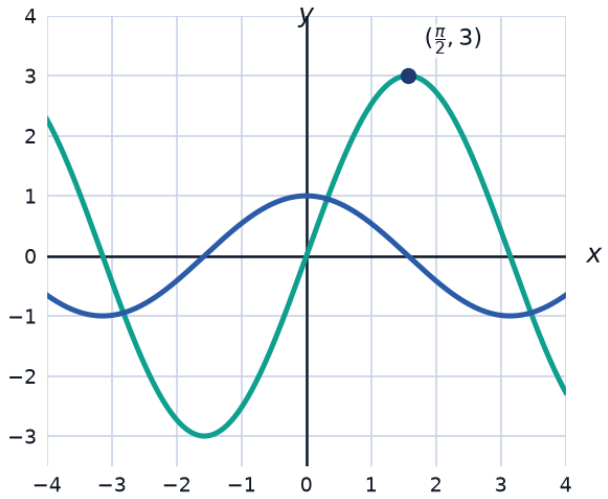
Solve the trigonometric equation on the interval $[0, 2\pi)$: $2\cos^2 x - 3\cos x + 1 = 0$

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all complex solutions to the cubic equation $z^3 + 8 = 0$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the limit: $\lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the scalar projection of $\mathbf{u} = \langle 4, 3 \rangle$ onto $\mathbf{v} = \langle 12, 5 \rangle$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Simplify the complex quotient into standard form $a + bi$: $\frac{3+4i}{1+2i}$

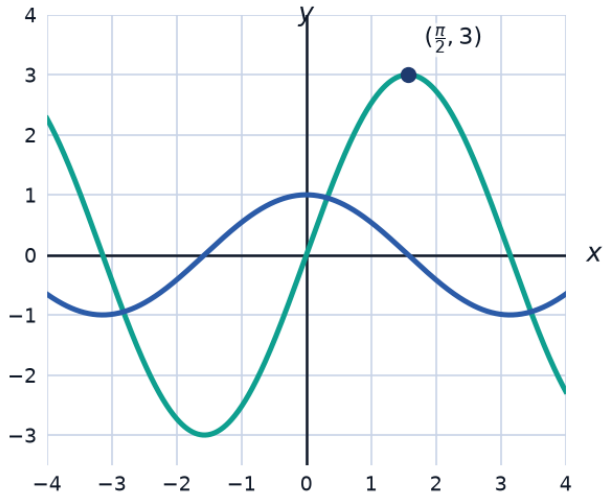
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the power of the imaginary unit: $i^{2026} + i^{2027} + i^{2028} + i^{2029}$

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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A constant force vector $\mathbf{F} = \langle 8, -3 \rangle$ (in newtons) moves an object along the displacement vector $\mathbf{d} = \langle 5, 2 \rangle$ (in meters). Calculate the work done by the force.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given that $\sin\theta = \frac{3}{5}$ with θ in Quadrant II, find the exact value of $\sin(2\theta)$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the indeterminate limit: $\lim_{x \to 5} \frac{x^2 - 25}{x^2 - 8x + 15}$

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, $A = 45^\circ$, $B = 60^\circ$, and $a = 8$. Find the exact length of side b.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve for x: $\log_2(x) + \log_2(x - 6) = 4$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

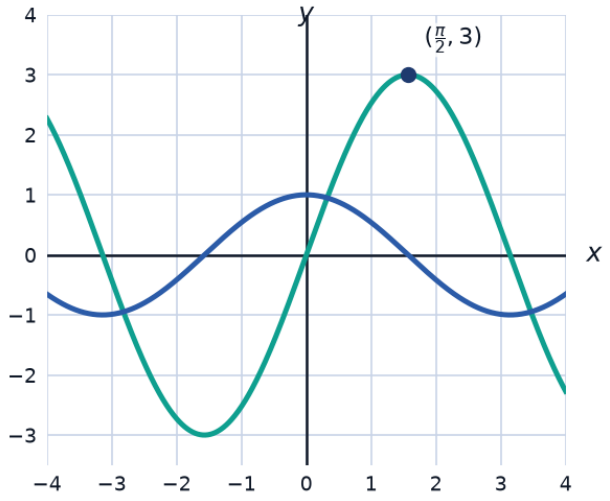
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Vectors $\mathbf{u} = \langle 3, -2, 5 \rangle$ and $\mathbf{v} = \langle 4, k, -2 \rangle$ are orthogonal in $\mathbb{R}^3$. Find the value of k.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

Evaluate the limit at infinity: $\lim_{x\to\infty} \frac{5x^3 - 2x + 7}{2x^3 + 4x^2 - 1}$



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Evaluate the limit: $\lim_{x\to 2} \frac{3x^2 - 4x + 1}{x + 3}$

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Solve the exponential equation for x: $4^{x+1} = 8^{2x-1}$

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

Given $A = (2, -1; 3, 4)$ and $B = (0, 5; -2, 1)$, compute $3A - 2B$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Use the Remainder Theorem to find the remainder when $P(x) = 3x^4 - 2x^3 + 5x - 7$ is divided by $x + 2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $2\sin(5x)\cos(3x)$ as a sum of trigonometric functions.

a. $\sin(8x) + \sin(2x)$

b. $\sin(8x) - \sin(2x)$

c. $\cos(8x) + \cos(2x)$

d. $\cos(8x) - \cos(2x)$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the expression $\frac{\sin(3x)+\sin(x)}{\cos(3x)+\cos(x)}$.

a. $\tan(2x)$

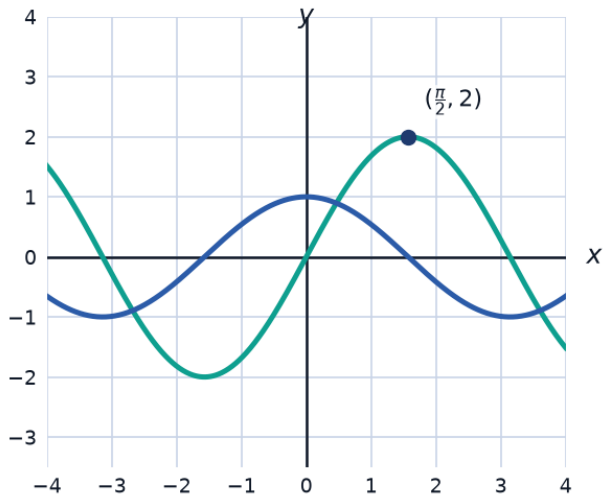
b. $\tan(x)$

c. $\cot(2x)$

d. $\tan(4x)$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the exact value of $\cos(105^\circ) - \cos(15^\circ)$?



a. $-\frac{\sqrt{2}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{6}}{2}$

d. $\frac{\sqrt{6}}{2}$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What geometric curve is represented by the polar equation $r = 2 + 2\cos\theta$?

a. Cardioid

b. Limaçon with inner loop

c. Three-petal rose

d. Circle

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3

b. 6

c. 4

d. 8

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 4

b. 8

c. 16

d. 2

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar point $(4, \frac{5\pi}{6})$ to rectangular coordinates (x, y).

a. $(-2\sqrt{3}, 2)$

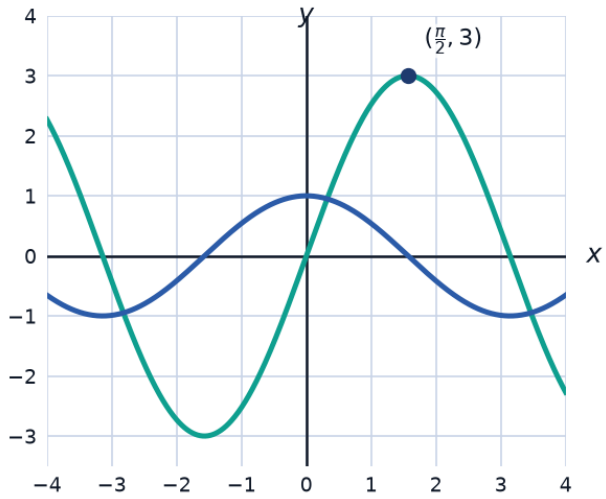
b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(2, -2\sqrt{3})$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular point $(-3, -3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.



a. $(3\sqrt{2}, \frac{5\pi}{4})$

b. $(3\sqrt{2}, \frac{3\pi}{4})$

c. $(6, \frac{5\pi}{4})$

d. $(3\sqrt{2}, \frac{7\pi}{4})$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the Cartesian equation for the polar line $\theta = \frac{\pi}{3}$?

a. $y = \sqrt{3}x$

b. $y = \frac{\sqrt{3}}{3}x$

c. $y = x + \frac{\pi}{3}$

d. $x = \sqrt{3}y$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $\cos(3x) = \frac{1}{2}$ for $x \in [0, \pi]$.

a. $\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{9}, \frac{7\pi}{9}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all solutions to $\sin(2x) = \cos(x)$ in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{6}, \frac{5\pi}{6}$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the range of $f(x) = 2 - 3\sin(4x)$?

a. $[-1, 5]$

b. $[-3, 3]$

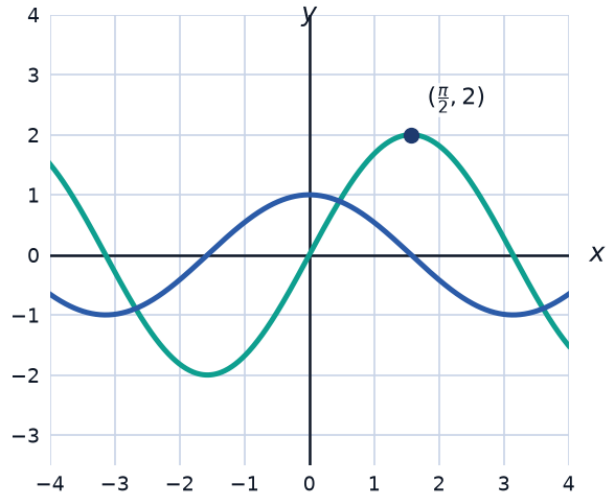
c. $[-5, 1]$

d. $[2, 5]$

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33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y)-coordinate plane, what are the rectangular coordinates of the point with polar coordinates $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$?



a. $(-3, 3\sqrt{3})$

b. $(-3\sqrt{3}, 3)$

c. $(3\sqrt{3}, -3)$

d. $(-3\sqrt{3}, -3)$

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A point in the polar coordinate plane has coordinates $(r, \theta) = \left(4, \frac{5\pi}{6}\right)$. Which of the following gives the rectangular coordinates (x, y) of this point?

a. $(-2\sqrt{3}, 2)$

b. $(-2, 2\sqrt{3})$

c. $(2, -2\sqrt{3})$

d. $(2\sqrt{3}, -2)$

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3.464, -2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4, 210^\circ)$

b. $(4, 30^\circ)$

c. $(4, 150^\circ)$

d. $(8, 210^\circ)$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3, -5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6, 240^\circ)$

b. $(6, 60^\circ)$

c. $(6, 120^\circ)$

d. $(12, 240^\circ)$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3.464,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 150^\circ)$

b. $(4,\ 330^\circ)$

c. $(4,\ 210^\circ)$

d. $(8,\ 150^\circ)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Convert the rectangular point $(3.464,\ -2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 330^\circ)$

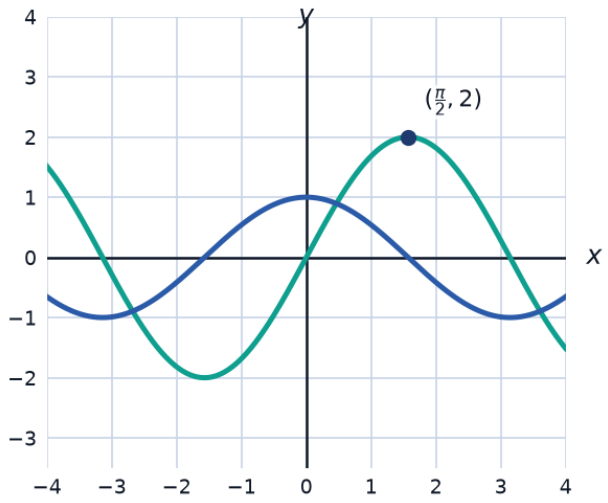
b. $(4,\ 150^\circ)$

c. $(4,\ 30^\circ)$

d. $(8,\ 330^\circ)$

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

Convert the rectangular point $(-3,\ 5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.



a. $(6,\ 120^\circ)$

b. $(6,\ 300^\circ)$

c. $(6,\ 240^\circ)$

d. $(12,\ 120^\circ)$

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Functions & Equations

Convert the rectangular point $(3,\ -5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6,\ 300^\circ)$

b. $(6,\ 120^\circ)$

c. $(6,\ 60^\circ)$

d. $(12,\ 300^\circ)$

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Limits

Convert the rectangular point $(-2,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(2.83,\ 135^\circ)$

b. $(2.83,\ 315^\circ)$

c. $(2.83,\ 225^\circ)$

d. $(6,\ 135^\circ)$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Complex Numbers

Convert the rectangular point $(3,\ 5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6,\ 60^\circ)$

b. $(6,\ 240^\circ)$

c. $(6,\ 300^\circ)$

d. $(12,\ 60^\circ)$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Convert the rectangular point $(3.464,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 30^\circ)$

b. $(4,\ 210^\circ)$

c. $(4,\ 330^\circ)$

d. $(8,\ 30^\circ)$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

Convert the rectangular point $(2,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(2.83,\ 45^\circ)$

b. $(2.83,\ 225^\circ)$

c. $(2.83,\ 315^\circ)$

d. $(6,\ 45^\circ)$

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Functions & Equations

Convert the rectangular point $(-4,\ 0)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 180^\circ)$

b. $(4,\ 0^\circ)$

c. $(4,\ 210^\circ)$

d. $(8,\ 180^\circ)$

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